INCH-POUND

MS21953C
w/AMENDMENT 1
14 February 2025
SUPERSEDING
MS21953C
9 April 2015

DETAIL SPECIFICATION SHEET

BODY, CLUSTER FITTING, ONE-WAY, FLARE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and SAE-AS4875/1.

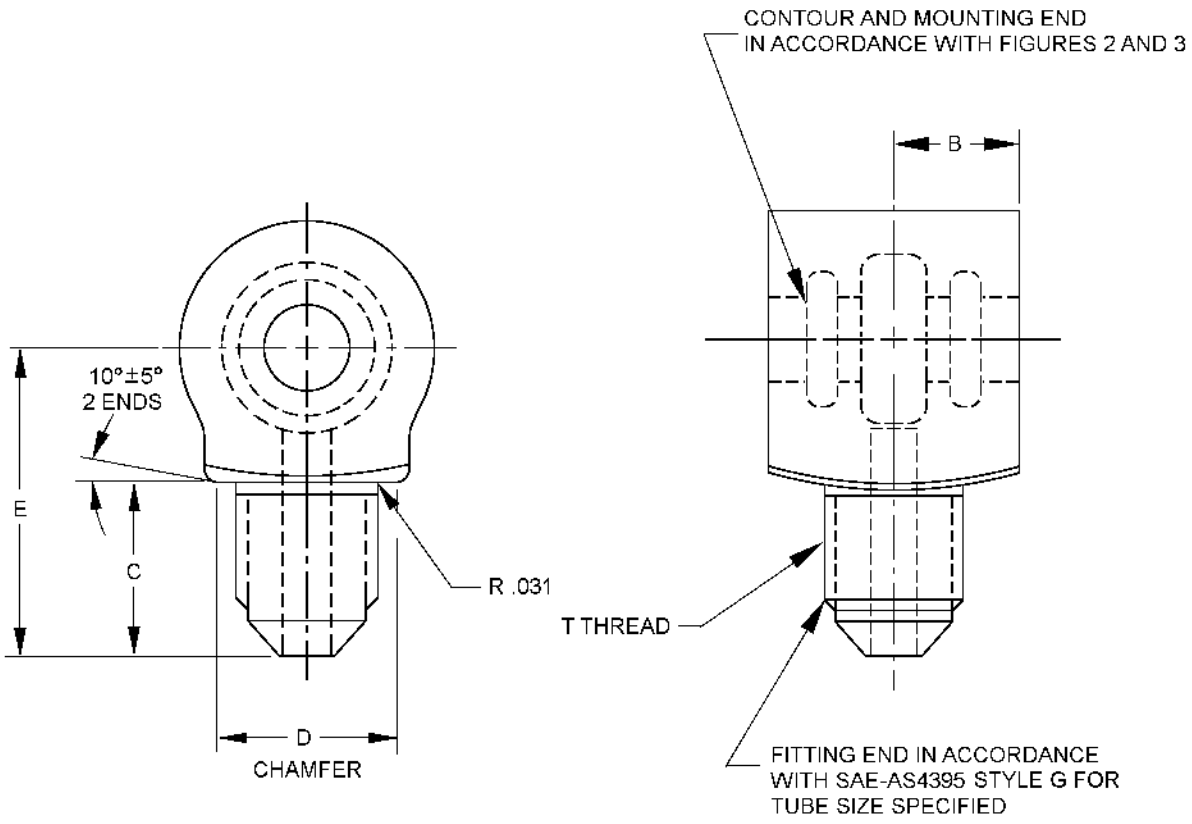


FIGURE 1. Body, cluster fitting.

AMSC N/A
DISTRIBUTION STATEMENT A: Approved for public release; Distribution is unlimited.

FSC 4730

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T14:42Z
Check the source to verify that this is the current version before use.

Dimensions for low flow sizes.

Low Flow Size						
Al Alloy 7075	Tube OD	Thread T SAE-AS8879	B	C	D	E
W3L	.187	.3750-24UNJF-3A	.500	.540	.710	1.109
W4L	.250	.4375-20UNJF-3A	.500	.625	.710	1.187
W5L	.312	.5000-20UNJF-3A	.500	.625	.710	1.187
W6L	.375	.5625-18UNJF-3A	.500	.640	.710	1.203

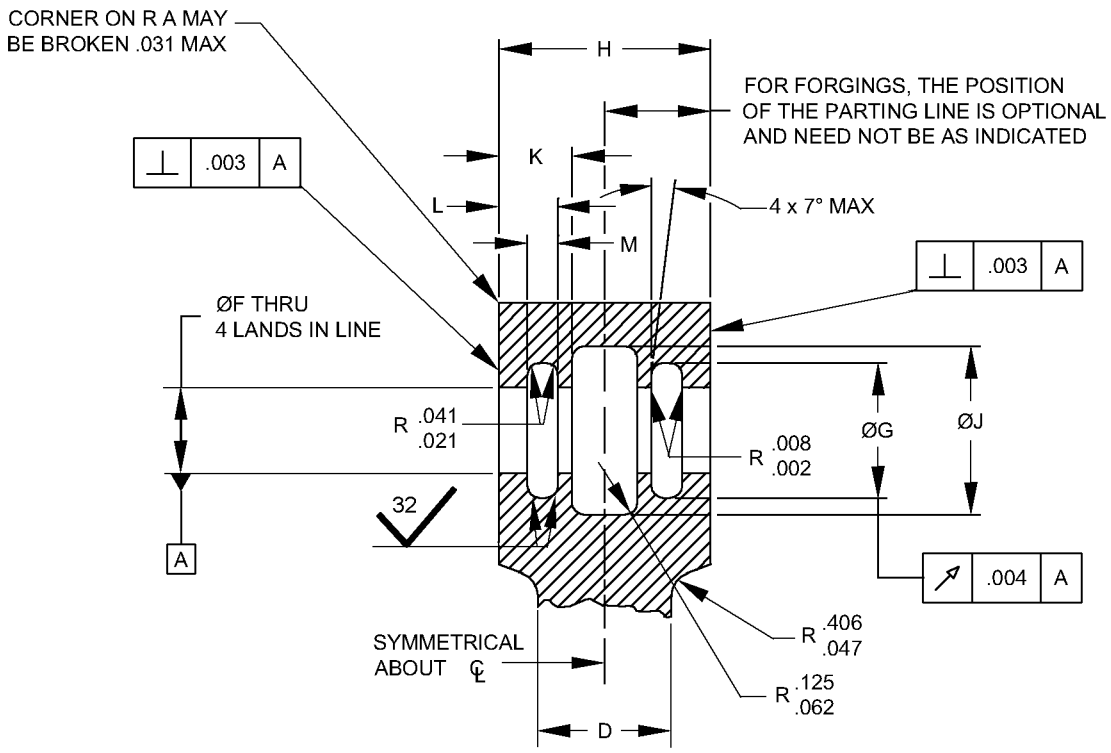
Dimensions for high flow sizes.

High Flow Size						
Al Alloy 7075	Tube OD	Thread T SAE-AS8879	B	C	D	E
W4H	.250	.4375-20UNJF-3A	.594	.625	.710	1.438
W5H	.312	.5000-20UNJF-3A	.594	.625	.710	1.438
W6H	.375	.5625-18UNJF-3A	.594	.648	.710	1.460
W8H	.500	.7500-16UNJF-3A	.594	.750	.750	1.562

NOTE: Dimensions are in inches. Unless otherwise specified, tolerances: decimals ± .010 inch, angles ± .5°.

FIGURE 1. Body, cluster fitting - Continued

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T14:42Z
Check the source to verify that this is the current version before use.



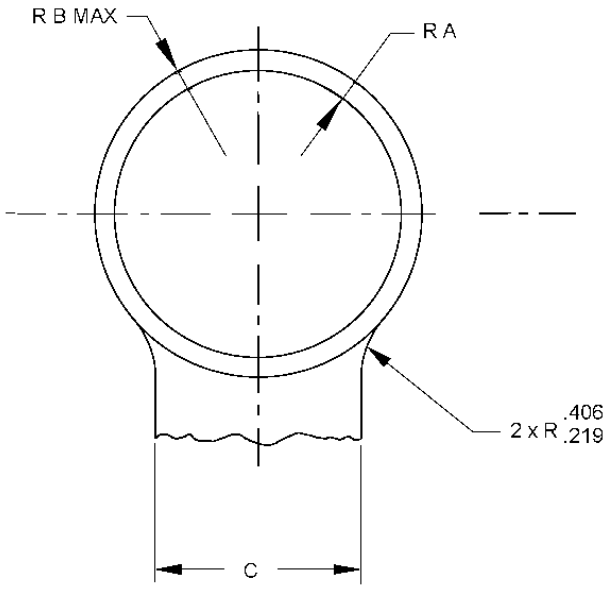
MOUNTING END DIMENSIONS

Type Fitting	D Max (see note 2)	F dia. ± .0005	G dia. ± .004	H ± .005	J dia. ± .010	K ± .010	L ± .004	M ± .005
Low Flow	.995	.5000	.664	1.000	.797	.312	.249	.143
High Flow	1.188	.8125	1.045	1.188	1.078	.375	.283	.175

NOTES:
1. Dimensions in inches. Unless otherwise specified, tolerances: decimals ± .010 inch, angles ± .5°.
2. Dimension D represents the max envelope which shall be large enough to meet final part dimensions.

FIGURE 2. Mounting end.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T14:42Z
Check the source to verify that this is the current version before use.



CONTOUR DIMENSIONS EXTERNAL

Type Fitting	A ± .031	B Max (see note 2)	C Max (see note 3)
Low Flow	.562	.636	.844
High Flow	.812	.909	1.156

NOTES:
1. Dimensions in inches. Unless otherwise specified, tolerances: decimals ± .010 inches.
2. B radius is the max envelope dimension for forging draft, flash, and other projections on A radius.
3. Dimension C represents the max envelope which shall be large enough to meet final part dimensions.

FIGURE 3. External contour.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T14:42Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

The requirements for acquiring the product described herein shall consist of this specification sheet and SAE-AS4875/1.

Dimensions. See figure 1, 2 and 3.

Dimensions and tolerancing in accordance with ASME Y14.5.

Fittings shall be free of all burrs and slivers.

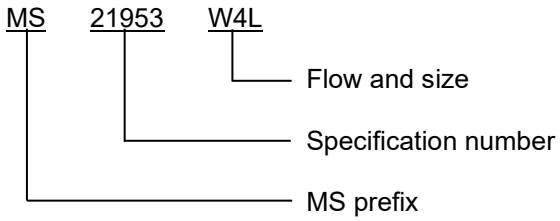
All machined surfaces shall be finished to 250 µin Ra, unless otherwise specified. Surface finish shall be in accordance with ASME B46.1.

Material: Aluminum alloy 7075-T73 or T7351 in accordance with SAE-AMS4124. Aluminum alloy forging 7075-T73 in accordance with SAE-AMS-QQ-A-367.

Finish: See acquisition standard SAE-AS4875/1. Aluminum alloy 7075 fittings shall be dyed brown.

Fittings shall be free of all burrs and slivers.

Part or Identifying Number (PIN): The PIN consists of the MS prefix, specification number, and flow and size (see figure 1).



Example of a PIN: MS21953W4L represents a low flow one-way body cluster flared fitting tube OD .250 inches.

Table I shows cancelled aluminum alloy 2014 and 2024 (-) PIN's which are replaced by aluminum alloy 7075 (W) PIN's.

TABLE I. Cancelled PIN to replacement PIN.

Cancelled PIN AL 2014 and 2024	Replacement PIN AL 7075
Low flow	
MS21953-3L	MS21953W3L
MS21953-4L	MS21953W4L
MS21953-5L	MS21953W5L
MS21953-6L	MS21953W6L
High flow	
MS21953-4H	MS21953W4H
MS21953-5H	MS21953W5H
MS21953-6H	MS21953W6H
MS21953-8H	MS21953W8H

Referenced documents shall be of the issue in effect on date of invitation for bid.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T14:42Z
Check the source to verify that this is the current version before use.

Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Reference documents. In addition to SAE-AS4875/1 this document references the following:

ASME B46.1
ASME Y14.5
SAE-AMS4124
SAE-AMS-QQ-A-367
SAE-AS4395
SAE-AS8879

CONCLUDING MATERIAL

Custodians:
Navy - AS
DLA - CC

Preparing activity:
DLA - CC

(Project 4730-2025-014)

Review activities:
Navy - SA

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T14:42Z
Check the source to verify that this is the current version before use.